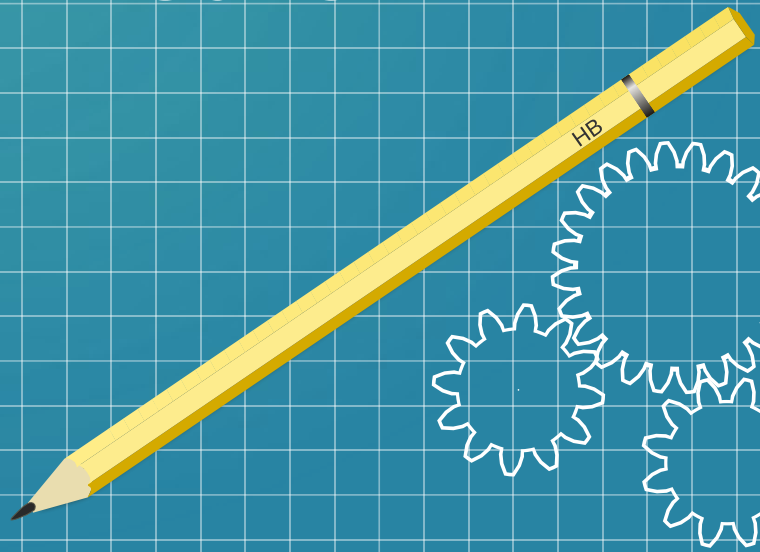
Emotional recognition from speech using CNN and four speech features

SIGNALS AND SYSTEMS PROJECT

2026-27

Nisarg Nayi
Yuvraj Singh
Ritesh Jaiswal
Prakhar Agrawal

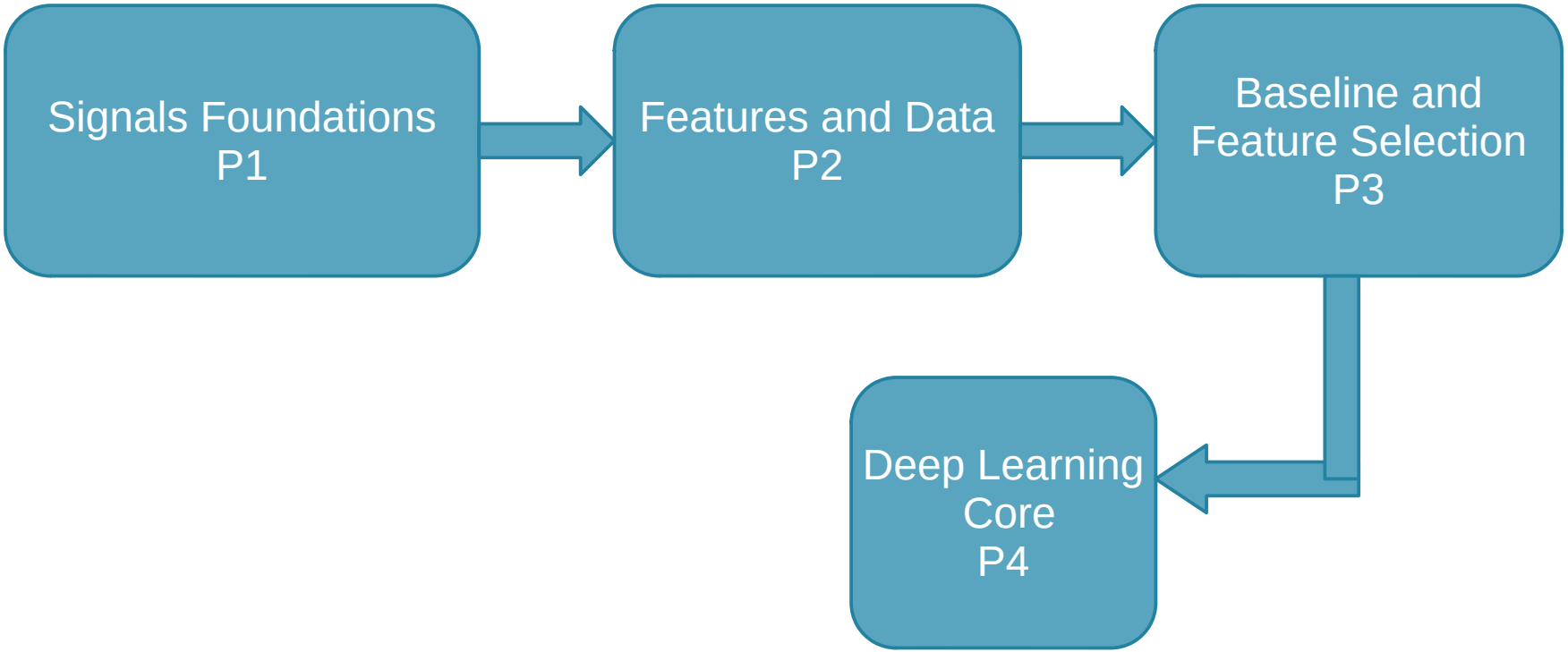


Litreature Survey

- Project aims to recognise human emotion by processing speech input using concepts of mathematics and signal processing
- Initial plan for implementation is applying concept of CNN as novelty and exploiting usage of MFCC,Sampling, Fast Fourier Transform, Framing, Windowing,
- Pre processing a dataset(RAVDESS) to train the output.

Initial Plan

- Project is planned to be implemented in 4 phases



Phase 1

Understanding audio signals

Concepts to Learn

- Audio signal; continuous vs discrete
- Amplitude, time-domain, waveform
- Sampling, sampling rate, sampling period

Phase 1

Understanding audio signals

Practical Work

- Load .wav files
- Visualize waveform
- See audio as numerical samples

Phase 1

Understanding audio signals

Tools

- Python
- NumPy
- Librosa
- Matplotlib

```
Voice → Microphone → Digital samples → Waveform
```

Phase 1

Frequency Analysis of Speech

Concepts to Learn

- Frequency; time vs frequency domain
- Fourier Transform, FFT
- Magnitude spectrum

- **Discrete Fourier Transform (DFT):** Converts a time-domain frame $x[n]$ into its component frequency bins $X[k]$:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot w[n] \cdot e^{-j\frac{2\pi}{N}kn}$$

- **Fast Fourier Transform (FFT):** An optimized algorithm calculating the DFT in $\mathcal{O}(N \log N)$ time using $N = 512$ Fourier coefficients.

- **Short-Time Fourier Transform (STFT):** Running the FFT across every consecutive frame yields a **2D Spectrogram** (Frequency bins $\times$ Time frames).

Phase 1

Frequency Analysis of Speech

Practical Work

- Waveform → FFT → Frequency spectrum
- Analyze dominant frequencies
- Energy distribution & speech components
-

Phase 1

Time Frequency Analysis

Concepts to Learn

- STFT, windowing, frames
- Spectrogram

Phase 1

Time Frequency Analysis

Practical Work

- Audio ⟶ STFT ⟶ Spectrogram
- Read axes: X = time, Y = frequency, colour = energy

Next Evaluation Milestone

This concludes Phase 1 Survey of the course project

Phase 2

- Data Set Creation
  - Surveying quality of data on RAVDESS and TESS to assess its usecase
- Speech Feature Extraction

  Using these learnt fundamentals of Signal Processing and implementing them on orignal dataset to get MFCC, Del MFCC, Spectogram and Chroma.